

Workshop 1 - Sentiment Analysis of Food Preferences for Event-Based Menu Recommendation Using NLP

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Abstract—Food selection is a critical component of event planning. This project proposes a data pipeline that integrates Natural Language Processing (NLP) with structured and unstructured data sources to analyze public opinion about food. The system aims to generate insights that support event-based menu recommendations using sentiment analysis, targeted web scraping and external food data APIs.

Index Terms—Sentiment Analysis, NLP, Data Pipeline, Food Recommendation, APIs, Web Scraping

I. INTRODUCTION

New York City stands as one of the most gastronomically diverse urban environments in the world, hosting thousands of restaurants and luxury hotels that regularly organize events ranging from corporate dinners to weddings and cultural celebrations. For catering managers and event planners, the ability to anticipate what dishes will resonate with specific audiences is both strategically and commercially critical.

Public opinion expressed through online platforms—particularly reviews, food blogs, and culinary forums—constitutes an underutilized source of insight for menu design. Structured recipe data from APIs like Spoonacular, when combined with unstructured opinion data, enables the construction of a data-driven recommendation layer that goes beyond generic popularity rankings.

This project addresses the challenge of transforming heterogeneous data sources into actionable menu intelligence by building an end-to-end NLP pipeline capable of extracting, processing, and visualizing food sentiment in the context of specific event types and NYC neighborhoods.

A. Project Overview

Title and Topic Angle: Sentiment analysis of food preferences for event-based menu recommendation.

Motivation: Understanding how the dining public perceives specific dishes, cuisines, and ingredients enables restaurants and hotel catering departments to make evidence-based decisions. In a competitive and trend-sensitive market like New York City, acting on real-time public sentiment can directly improve guest

satisfaction and event outcomes. This pipeline provides a scalable mechanism to operationalize that insight continuously.

Goal: To design a data engineering and NLP pipeline capable of collecting, processing, and analyzing food-related opinions to support menu recommendation.

Scope and Limitations:

- **Scope:** Integration of API and web data, sentiment analysis, and trend extraction.
- **Limitations:** API rate limits, noisy user-generated content, and limited contextual understanding of events.

II. SOURCE CHARACTERIZATION

A. API Source — Spoonacular

Spoonacular is a food and recipe API that provides structured data about recipes, ingredients, and nutritional information. It was selected as the primary structured data source because it offers direct access to culinary metadata that forms the catalog of dishes the recommendation system evaluates.

Characteristics:

- REST API with JSON responses; authentication via API key.
- Endpoint used: `/recipes/findByIngredients`, queried through the wrapper function `collect_recipes_by_ingredient`, which accepts the following parameters:
 - `ingredient` — main ingredient to search for.
 - `number` — number of recipes to retrieve (default: 10).
 - `intolerances` — dietary intolerances to filter out (e.g., *gluten*, *dairy*).
 - `diet` — dietary preference filter (e.g., *vegan*, *vegetarian*).
- Returns structured records including recipe title, matched ingredients, missing ingredients, and recipe ID for further lookup.

Relevance: This API provides the structured ingredient and recipe layer of the pipeline — the “what

a dish is” side of the recommendation engine. The intolerances and diet parameters are particularly relevant for event-based menu planning, as they allow filtering dishes to match the dietary needs of specific guest profiles directly at ingestion time.

Validation: A successful request returns a JSON array where each object includes `id`, `title`, `usedIngredientCount`, `missedIngredientCount`, and `missedIngredients`. A representative sample of 20 raw records is included in the `workshop_1/` directory of the repository.

B. Web Scraping Source — Eater NY (RSS Feed)

Eater NY is a leading food blog and news publication dedicated to the New York City culinary scene. Initially, platforms like TripAdvisor were considered; however, strict Web Application Firewalls (WAF) prevented ethical, automated extraction. Consequently, the Eater NY RSS feed was selected as the primary unstructured data source because it provides highly reliable, opinion-rich, and public-driven insights into emerging food trends and popular dishes in NYC without triggering anti-bot mechanisms.

Characteristics:

- **Opinion-rich text:** Article summaries and descriptions contain descriptive text about trending dishes, ingredients, and culinary expectations — essential for NLP, sentiment analysis, and entity extraction.
- **Reliable Delivery (XML/RSS):** Utilizing the native RSS feed bypasses aggressive DOM changes and WAF restrictions common in standard HTML scraping, ensuring a stable and ethical ingestion process.
- **Structured Metadata:** Articles are accompanied by pre-assigned categories (e.g., "NYC Restaurant Openings", "NYC Restaurant News") that allow targeted filtering of event-relevant dining trends.
- **Fields to extract:** article title, URL, author, article summary (raw text), categories, publication date, and extraction timestamp.

Relevance: Eater NY complements Spoonacular by providing the “market demand” layer — identifying what dishes or ingredients are currently trending in the NYC food scene. Together, both sources enable the pipeline to bridge unstructured public interest with structured recipe metadata, forming the basis of the event-driven recommendation engine.

III. EXTRACTION STRATEGY

A. Ingestion Periodicity

Both sources follow a **weekly schedule**, with an additional **event-driven trigger** activated whenever a new event is registered in the system (e.g., a corporate dinner, wedding, or private gathering).

- **Spoonacular API:** Weekly. Recipe catalogs and ingredient metadata are relatively stable, so daily ingestion would offer minimal new information at the cost of API quota. When an event is registered,

an additional on-demand run is triggered with the `diet` and `intolerances` parameters adjusted to match the guest profile of that event.

- **Eater NY (RSS Feed):** Weekly. Culinary trends, restaurant news, and public opinions evolve gradually; weekly snapshots of article summaries are sufficient to capture sentiment shifts relevant to menu planning. Event triggers also apply here to refresh the latest gastronomic news corpus before a specific event.

Periodicity justification: Unlike fast-moving topics such as sports or breaking news, food sentiment in established NYC restaurants changes at a slower cadence. Weekly ingestion balances data freshness with resource efficiency. The event-driven layer ensures that when a planner needs up-to-date recommendations for a specific occasion, the pipeline can be re-run on demand without altering its schedule.

B. Raw Data Format

All ingested data is persisted as **JSON files** with datetime-based naming to enable version control and historical traceability:

- **Spoonacular:**
`spoonacular_recipes_YYYYMMDD_HHMMSS.json`
- **Eater NY (RSS Feed):**
`eaterny_trends_YYYYMMDD_HHMMSS.json`
Files are stored in `datalake_bronze/` without any transformation applied at ingestion time.

C. Metadata Fields

The following metadata fields are persisted alongside the raw text content of each record to ensure data quality and traceability:

- `source` — origin platform (Spoonacular).
- `timestamp` — datetime of ingestion (YYYY-MM-DD HH:MM:SS).
- `author` — reviewer username or recipe author.
- `date` — original publication date of the record.
- `url` — source URL for traceability and re-validation.
- `language` — detected language of the text content.
- `category` — cuisine type or restaurant category.
- `event_keywords` — event-related terms detected in the record (e.g., *wedding*, *corporate*, *catering*).

IV. FUNCTIONAL USER DEFINITION

The primary functional user is a **Hotel or Restaurant Catering Manager / Event Planner** operating in New York City.

Profile:

- Medium technical knowledge—comfortable with web dashboards but not with raw code.
- Requires actionable insights: top-rated dishes per event type, trending ingredients, dietary demand signals.
- Makes weekly or per-event decisions on menu curation and event proposals.

- Operates in a competitive market where guest satisfaction is a primary KPI.

Justification: This user benefits from the pipeline by replacing ad-hoc intuition with data-backed menu choices. Visualizing which dishes receive the highest positive sentiment for weddings, corporate events, or private parties enables confident, client-facing proposals aligned with current NYC dining trends.

V. USER STORIES

User stories are prioritized by importance for dashboard design:

- 1) **[High]** As a catering manager, I want to see the highest-rated dishes for a specific type of event so I can offer clients a competitive menu.
- 2) **[High]** As an event planner, I want to filter dish suggestions by dietary restriction (vegan, gluten-free, halal) so that I can accommodate all guest profiles in a single proposal.
- 3) **[High]** As a restaurant manager, I want to monitor sentiment trends over time for specific dish categories so that I can detect which cuisines are gaining or losing popularity in NYC.
- 4) **[Medium]** As a data analyst, I want to view data quality metrics (missing fields, ingestion success rate, record counts) so that I can ensure the pipeline produces reliable inputs for the recommendation engine.
- 5) **[Medium]** As an event planner, I want to identify dishes frequently mentioned alongside specific NYC neighborhoods so that I can tailor menus to location-specific audiences.
- 6) **[Low]** As a manager, I want to receive automated alerts when a dish category experiences a sudden sentiment drop so that I can proactively remove it from upcoming event menus.
- 7) **[Low]** As a data analyst, I want to compare sentiment distributions between API-sourced recipe metadata and reviews so that I can validate the consistency of the recommendation model.

VI. PIPELINE ARCHITECTURE OVERVIEW

The system follows the **Medallion Architecture**, organizing data into three progressive layers as shown in Figure 1.

A. Layer Descriptions

A. Bronze — Raw Ingestion: Two **Apache Airflow** DAGs handle data collection: one queries the Spoonacular API via Python `requests`, and a second ingests data from the web source to be defined. Airflow was chosen for its retry logic, failure alerts, and monitoring interface. All responses are stored as timestamped JSON files in `datalake_bronze/` without transformation.

B. Silver — Cleaning and NLP: An Airflow **FileSensor** detects new Bronze files and triggers the Silver DAG automatically. **Pandas** handles data cleaning and normalization. The NLP pipeline applies entity extraction, sentiment scoring via **VADER** — suited for short-form opinion text — and keyword and topic extraction. Output is stored as **Parquet** in `datalake_silver/`, offering significant compression and faster analytical reads compared to JSON.

C. Gold — Aggregations: **Pandas** computes daily summaries: top dishes by sentiment per event type and trend windows. **PySpark** is available as a scale-out option if data volume grows beyond memory limits.

D. Dashboard: **Plotly Dash** reads the Gold summaries and renders an interactive dashboard accessible via browser. All components are containerized with **Docker** and launched via a single `docker-compose.yml`.

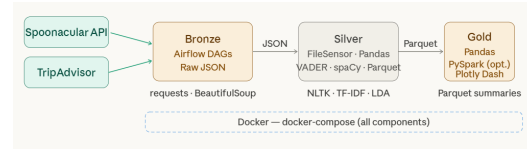


Fig. 1. Medallion pipeline.

B. Orchestration Strategy

The pipeline is orchestrated using **Apache Airflow**, a platform that allows defining workflows as **DAGs** (Directed Acyclic Graphs). A DAG is a Python script that specifies a set of tasks and the order in which they must execute — each task only runs after the previous one completes successfully, and the flow never loops back. Airflow provides a web interface to monitor task status, inspect logs, and configure automatic retries on failure.

The pipeline uses two DAGs:

- 1) **DAG 1 — Bronze ingestion** (scheduled daily at 00:00 UTC). This DAG runs on a fixed schedule and executes three tasks in sequence: call the Spoonacular API, call the web data source, and persist both responses as timestamped JSON files in `datalake_bronze/`. If any task fails, Airflow retries it automatically and sends an alert.
- 2) **DAG 2 — Silver processing** (event-driven). Instead of running on a fixed schedule, this DAG starts with a **FileSensor** — a special Airflow task that monitors `datalake_bronze/` and waits until new files appear. The moment DAG 1 writes new data, the sensor triggers the rest of the pipeline automatically: data cleaning with **Pandas**, NLP processing (entity extraction, sentiment scoring, topic modelling), and finally writing the results as **Parquet** files to `datalake_silver/`.

Once DAG 2 completes, a **Gold aggregation task** runs downstream: it reads the Silver Parquet files, com-

puts the daily summaries (top dishes by sentiment, trend windows, dietary frequencies), and writes the final outputs to `datalake_gold/`. These aggregated files are immediately consumed by the Plotly Dash dashboard.

VII. DATA SAMPLE AND VALIDATION

A. Web Scraping Sample (Eater NY RSS Feed)

A sample dataset was extracted from the Eater NY RSS feed to validate data accessibility and structure, successfully bypassing the strict Web Application Firewalls (WAF) encountered during standard HTML scraping. The dataset provides unstructured text essential for NLP and entity extraction, capturing fields such as article titles, detailed summaries, publication dates, and gastronomic categories. A fragment of the collected data is shown below:

```
1 [
2   {
3     "source": "Eater NY (Atom)",
4     "article_title": "Fashion-y Hangout
      Jeans Will Expand to the
      Hamptons",
5     "article_url": "https://ny.eater.com/
      restaurant-openings/410069/jeans-
      noho-hamptons-chodorow-ashwin-
      deshmkh",
6     "article_id": "https://ny.eater.com/?
      p=410069",
7     "author": "Beth Landman",
8     "article_summary": "The team behind
      popular NYC spot Jean's has
      purchased the former Old Stove
      Pub in Sagaponack for $4.5
      million. Led by Jeffrey Chodorow,
      the Hamptons expansion will
      feature a seafood-focused, family
      -friendly menu and is expected to
      open in 2027, despite a current
      naming dispute with the previous
      owner.",
9     "categories": [
10      "NYC Restaurant Openings"
11    ],
12     "location_focus": "New York City",
13     "published_date": "2026-03-20T17:59:0
      3-04:00",
14     "date_extracted": "20260322_105127"
15   },
16   {
17     "source": "Eater NY (Atom)",
18     "article_title": "An East Village
      Pizzeria and More NYC
      Restaurant Closures in March",
19     "article_url": "https://ny.eater.com/
      news/409812/nyc-restaurants-
      closed-march-2026",
20     "article_id": "https://ny.eater.com/?
      p=409812",
21     "author": "Tierney Plumb",
22     "article_summary": "A roundup of NYC
      restaurant closures in March 2026,
      including the merger of East
      Village pizzeria Gruppo with
      Spunto, the closure of Park Slope's
      Everyman Espresso due to heavy
      competition, and the shuttering
      of the Upper West Side's Jing
      Fong and Chelsea's Elmo, among
      several others across the
      boroughs.",
```

```
      "categories": [
        "NYC Restaurant Closings",
        "NYC Restaurant News"
      ],
      "location_focus": "New York City",
      "published_date": "2026-03-20T13:02:3
        3-04:00",
      "date_extracted": "20260322_105127"
    },
    {
      "source": "Eater NY (Atom)",
      "article_title": "No Slowdown for the
        New York Slice",
      "article_url": "https://ny.eater.com/
        restaurant-openings/410057/no-
        slowdown-for-the-new-york-slice",
      "article_id": "https://ny.eater.com/?
        p=410057",
      "author": "Melissa McCart",
      "article_summary": "New York's slice
        shop scene continues to expand. L
        'Industrie has opened a third
        location in Little Italy next to
        Ferrara, See No Evil Pizza is
        launching a new slice format in
        Greenwich Village, and Paulie Gee
        's is adding two employee-owned
        spots: a slice shop in the East
        Village and a tavern in Gowanus."
      ,
      "categories": [
        "NYC Restaurant Openings"
      ],
      "location_focus": "New York City",
      "published_date": "2026-03-19T17:19:4
        3-04:00",
      "date_extracted": "20260322_105127"
    }
  ]
```

B. API Sample (Spoonacular API)

A sample dataset was retrieved from the Spoonacular API to validate the extraction of structured culinary data that will be paired with the trending dishes identified in the scraping phase. The dataset includes detailed recipe information, such as dish names, required ingredients, dietary classifications, and preparation metrics. A fragment of the collected data is shown below:

```
{
  "id": 640352,
  "title": "Cranberry Apple Crisp",
  "image": "https://img.spoonacular.com
    /recipes/640352-312x231.jpg",
  "imageType": "jpg",
  "usedIngredientCount": 1,
  "missedIngredientCount": 3,
  "missedIngredients": [
    {
      "id": 9078,
      "amount": 2.0,
      "unit": "cups",
      "unitLong": "cups",
      "unitShort": "cup",
      "aisle": "Produce",
      "name": "cranberries",
      "original": "2 cups fresh
        cranberries",
      "originalName": "fresh
        cranberries",
      "meta": [
        "fresh"
      ]
    }
  ]
}
```

```

22     },
23     "extendedName": "fresh
        cranberries",
24     "image": "https://img.
        spoonacular.com/
        ingredients_100x100/
        cranberries.jpg"
25 },
26 {
27     "id": 1145,
28     "amount": 4.0,
29     "unit": "Tbs",
30     "unitLong": "Tbs",
31     "unitShort": "Tbsp",
32     "aisle": "Milk, Eggs, Other
        Dairy",
33     "name": "butter",
34     "original": "1/2 stick (4 Tbs
        ) unsalted butter, cut
        into cubes",
35     "originalName": "1/2 stick
        unsalted butter, cut into
        cubes",
36     "meta": [
37         "unsalted",
38         "cut into cubes"
39     ],
40     "extendedName": "unsalted
        butter",
41     "image": "https://img.
        spoonacular.com/
        ingredients_100x100/
        butter-sliced.jpg"
42 },
43 {
44     "id": 8120,
45     "amount": 1.5,
46     "unit": "cups",
47     "unitLong": "cups",
48     "unitShort": "cup",
49     "aisle": "Cereal",
50     "name": "regular oats",
51     "original": "1 1/2 cups
        regular oats (not quick-
        cooking)",
52     "originalName": "regular oats
        (not quick-cooking)",
53     "meta": [
54         "(not quick-cooking)"
55     ],
56     "image": "https://img.
        spoonacular.com/
        ingredients_100x100/
        rolled-oats.jpg"
57 }

```

VIII. RESULTS & DISCUSSION

Initial validation confirms that both sources successfully provide sufficient data for sentiment and entity analysis. The Spoonacular API offers highly structured culinary data, while the Eater NY RSS feed provides rich, unstructured textual opinions on market trends. However, a preliminary data quality assessment revealed specific preprocessing challenges that must be addressed in the Silver layer pipeline:

- **Eater NY (Scraping Noise):** The extracted article summaries contain significant structural noise. The raw text includes embedded HTML entities (e.g., `’`, ` `), journalistic boilerplate, and photo credits (e.g., “Photo by Patrick McMullan via Getty Images”). This unstructured data will

require aggressive cleaning and text normalization using regular expressions before NLP models can accurately process the sentiment.

- **Spoonacular API (Nested Structures):** While the API data is clean, it is heavily nested and contains operational metadata unnecessary for NLP. The transformation process will require flattening nested JSON arrays (such as iterating through `missedIngredients` to extract only the name or `originalName` fields), standardizing complex ingredient strings (e.g., parsing “1/2 stick (4 Tbs) unsalted butter, cut into cubes” into core entities), and dropping non-textual fields like image URLs and numerical IDs to optimize storage and processing speed.

Note: JSON snippets demonstrating successful raw data extraction from both the Spoonacular API and the Eater NY RSS feed are included in Section VII of this report to serve as the proof of concept.

IX. REPOSITORY

The full project source code, data samples, and documentation are available at the following GitHub repository:

[https://github.com/vydidot/
Food-Gastronomy-Trends-Sentiment](https://github.com/vydidot/Food-Gastronomy-Trends-Sentiment)

X. CONCLUSIONS

This workshop establishes the conceptual and technical foundation for an NLP-driven menu recommendation pipeline targeting event planners and catering managers in New York City. Two complementary data sources were identified and validated: the **Spoonacular API**, which provides structured recipe and ingredient metadata, and **TripAdvisor**, which supplies opinion-rich user reviews from NYC dining establishments.

The combination of both sources enables the pipeline to bridge the gap between “what a dish is” (structured API data) and “how people feel about it” (unstructured review text), forming a solid basis for event-aware menu recommendations.

The proposed Medallion architecture (Bronze → Silver → Gold) ensures a clean separation between raw ingestion, NLP processing, and aggregated insights, while Apache Airflow provides reliable orchestration through weekly and event-driven DAG triggers.

Future work will focus on implementing the Bronze DAGs, completing the 20-record data samples from both sources, developing the Silver NLP pipeline (entity extraction, VADER sentiment scoring, TF-IDF keyword extraction), and deploying the initial Plotly Dash prototype.

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